

3. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$S \rightarrow 0A1$ $A \rightarrow 0A \mid 1A \mid \epsilon$

Code:

```
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    printf("Enter string: ");
    scanf("%s", s);

    int n = strlen(s);

    if (n >= 2 && s[0] == '0' && s[n - 1] == '1')
        printf("String belongs to the CFG");
    else
        printf("String does not belong to the CFG");

    return 0;
}
```

Output:

<pre>#include <stdio.h> #include <string.h> int main() { char s[100]; printf("Enter string: "); scanf("%s", s); int n = strlen(s); if (n >= 2 && s[0] == '0' && s[n - 1] == '1') printf("String belongs to the CFG"); else printf("String does not belong to the CFG"); return 0; }</pre>	<pre>Enter string: 001 String belongs to the CFG === Code Execution Successful ===</pre>
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